



Changes in cardiac function before and after hot water immersion and endurance exercise in middle-aged individuals with overweight or obesity

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Received: 19 February 2026 / Accepted: 23 August 2026
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Abstract

Purpose We investigated the acute effects of ecologically valid hot water immersion (HWI) and endurance exercise (END) interventions on indices of left (LV) and right ventricular (RV) function in middle-aged individuals with overweight or obesity.

Methods Twenty-five (15♀, 10♂) individuals were recruited. A single session of HWI (40 °C for 30 min with the water level at the sternum) and END (cycling 30 min at 60–70% of max heart rate [HR]) were carried out in randomised order, separated by at least 72 h. Core body temperature (Tc) and HR were monitored throughout each session. Blood pressure and ventricular parameters, including LV global longitudinal strain (LV GLS) and tricuspid annular plane systolic excursion (TAPSE), were assessed before and after each session.

Results A significant increase in Tc was greater after HWI ($\Delta 0.91 \pm 0.32$ °C) than END ($\Delta 0.72 \pm 0.29$ °C; interaction: $p = 0.004$). Similarly, HR increased more after HWI than END ($\Delta 33 \pm 14$ vs. $\Delta 25 \pm 12$ bpm; $p = 0.042$) and HWI induced greater reductions in post-intervention SBP ($\Delta -5.7 \pm 10.0$ vs. $\Delta 0.6 \pm 6.6$ mmHg; $p = 0.019$) and DBP ($\Delta -9.2 \pm 9.2$ vs. $\Delta -1.5 \pm 5.1$ mmHg; $p = 0.001$). There was a greater increase in LV GLS following HWI ($\Delta -2.5 \pm 2.3\%$) compared to END ($\Delta -0.76 \pm 2.09\%$; $p = 0.008$) and a significant increase in TAPSE after HWI ($\Delta 3.95 \pm 4.32$ mm; $p < 0.001$).

Conclusion We observed greater increases in Tc, HR, and post-exercise LV and RV function following a HWI intervention compared to acute END. The physiological and practical importance of these differences should be explored further in long-term chronic programs in populations that could benefit from an alternative to the traditional END for cardiovascular benefits.

Communicated by Ting-Heng Chou.

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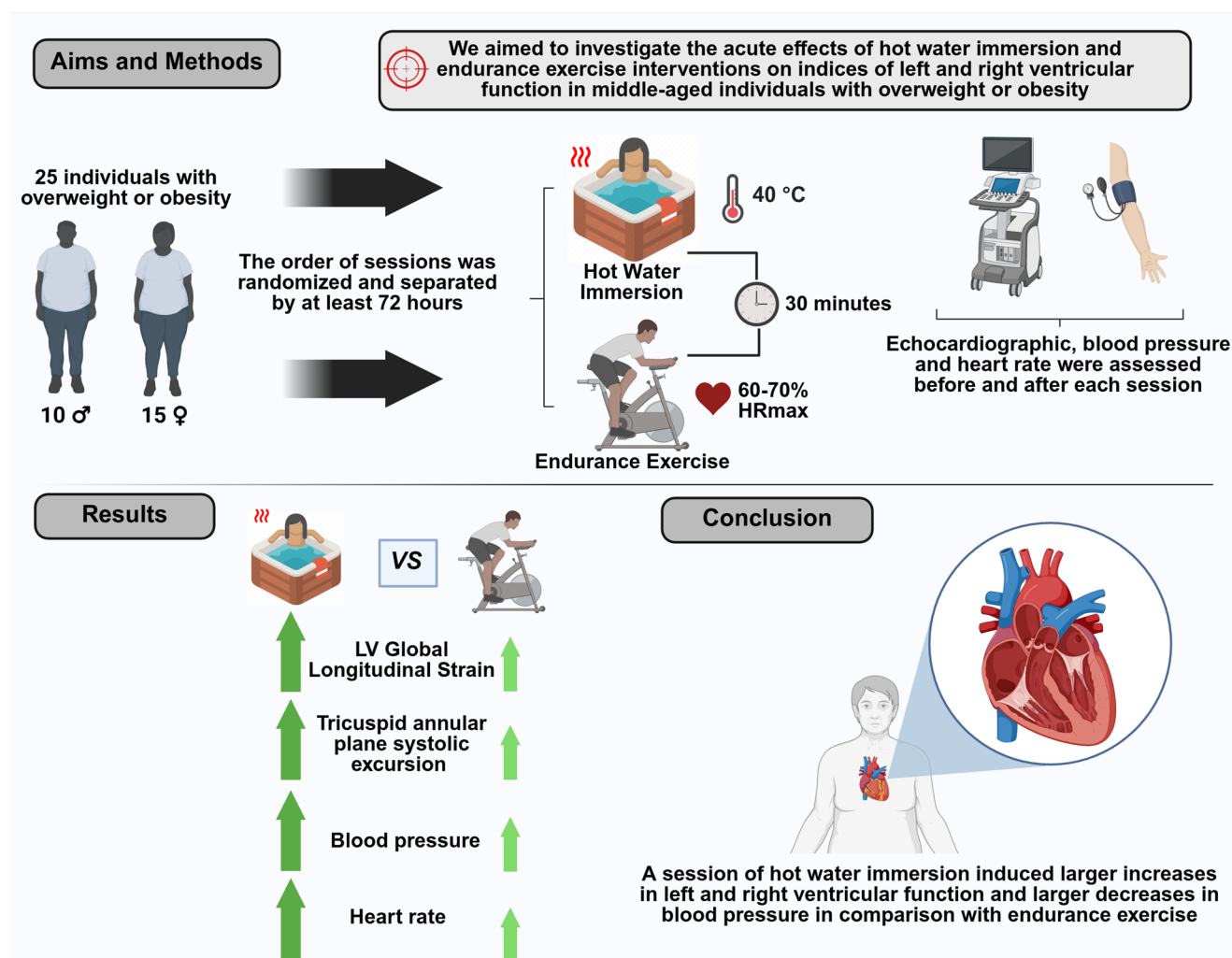
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Graphical abstract



Keywords Heat stress · Left ventricle · Right ventricle · Global longitudinal strain · Aerobic exercise

Abbreviations

A	Late transmitral inflow velocity
a'	Annulus tissue velocity in late diastole
ANOVA	Analysis of variance
BMI	Body mass index
DBP	Diastolic blood pressure
DXA	Dual-energy X-ray absorptiometry
E	Early transmitral inflow velocity
e'	Annulus tissue velocity in early diastole
EDV	End-diastolic volume
EF	Ejection fraction
END	Moderate-intensity endurance exercise
ESV	End-systolic volume
GLS	Global longitudinal strain
HR	Heart rate
HWI	Hot water immersion

IPAQ	International Physical Activity Questionnaire
LV	Left ventricle
MAP	Mean arterial pressure
MV	Mitral valve
Q	Cardiac output
RV	Right ventricle
RVFWSL	Right ventricular free wall longitudinal strain
s'	Systolic velocity
SBP	Systolic blood pressure
SPSS	Statistical Package for the Social Sciences
SV	Stroke volume
TAPSE	Tricuspid annular plane systolic excursion
Tc	Body core temperature
VO ₂ max	Maximal aerobic capacity

Introduction

Aging and excess body weight are associated with decreased thermoregulatory capacity and impaired cardiac function (Morrissey et al. 2021; Franceschi et al. 2018). Older overweight and obese individuals have difficulty performing exercise at levels that may provide cardiovascular adaptation and benefit. Recent studies have suggested that the beneficial cardiovascular effects of hot water immersion (HWI) are similar to, or greater than, the well-established effects of exercise (Brunt and Minson 2021). Akerman et al. 2019 demonstrated a reduction of systolic blood pressure (SBP) of 7 mmHg following 12 weeks of HWI combined with calisthenics in elderly individuals with peripheral artery disease, which exceeded the 4 mmHg decrease following exercise alone (Akerman et al. 2019). A recent meta-analysis revealed that sauna-based heat therapy improved left ventricular (LV) ejection fraction (EF) by an average of 3.27% (Li et al. 2020). HWI may represent a non-pharmacological alternative approach for individuals with obesity who are unable to perform regular endurance exercise at a frequency and intensity that induces beneficial effects on cardiac function, but a paucity of data currently exists regarding the acute effects of HWI versus exercise in older individuals with overweight and/or obesity.

Acute heat stress may be an effective intervention for inducing changes in cardiac function. A recent meta-analysis of over 400 laboratory-based heat exposure studies revealed that acute exposure to heat stress resulted in significant elevations in heart rate (HR) and cardiac output (Q) (Meade et al. 2025). This is in agreement with previous evidence that demonstrated a reduction in central blood volume following passive body heating, alongside significant compensatory increases in ejection fraction (EF) and HR (Crandall et al. 2008). Furthermore, previous evidence revealed that an increase in core body temperature of $\sim 1^\circ\text{C}$ significantly increased ventricular systolic function by augmenting septal and lateral mitral annular tissue velocities (Brothers et al. 2009). Despite this body of evidence, the effects of HWI on more sensitive parameters of cardiac function, such as left ventricle global longitudinal strain (LV GLS), are unknown. This measure is a sensitive marker of LV function capable of detecting early preclinical changes in systolic function (Karlsen et al. 2019).

Few studies have directly compared the acute effects of HWI and exercise on cardiovascular outcomes. Amin and colleagues (2022) compared the effects of a 30-minute HWI session to a bout of moderate- and high-intensity exercise, and all 3 interventions promoted similar increases in femoral artery shear rate. Thomas and colleagues (2016) compared a time-matched 30-minute session of lower-limb HWI (42°C) to treadmill running (65–75% of age-predicted

HRmax) in 10 healthy young adults (8 males and 2 females), demonstrating that only exercise significantly increased Q. Furthermore, a time-matched comparison between 60 min of HWI at 40.5°C versus cycling at 60% of maximal aerobic capacity ($\text{VO}_{2\text{max}}$) found that exercise elicited significantly greater increases in HR, Q and mean arterial pressure (MAP). Moreover, exercise also induced a significant reduction in stroke volume (SV) (Francisco et al. 2021). These studies comparing the acute effects of HWI and exercise had relatively small sample sizes, and recruited young healthy individuals. Individuals with obesity are at an increased risk of presenting subclinical cardiac dysfunction (Chen et al. 2023) and might present difficulties in performing weight-bearing exercises. Additionally, despite evidence on vascular outcome measures, only 2 studies have included cardiac function in their analyses, and neither assessed contemporary echocardiographic indices, such as GLS, and indices of right ventricular (RV) function, despite their clinical relevance and predictive value for cardiovascular diseases and all-cause mortality (Amsallem et al. 2018).

This study compared the acute effects of standard, ecologically-valid prescriptions for HWI and moderate-intensity endurance exercise (END) on cardiac function in middle-aged individuals with overweight/obesity.

Methods

Study design

This is a sub-study from a clinical trial, which was prospectively registered in the Australian New Zealand Clinical Trials Registry (ACTRN12623000724673) and approved by the local human ethics committee (2022/ET000738). In addition, this study accorded with the 1964 Declaration of Helsinki and its later amendments. Recruitment was performed via digital advertisement on social media and targeted middle-aged individuals with overweight and obesity. The inclusion criteria included: (1) age between 45 and 70 years; (2) Body mass index (BMI) $\geq 27\text{ kg/m}^2$ with the addition of an obesity-related comorbidity; or BMI $\geq 30\text{ kg/m}^2$; (3) sedentary activity levels according to the International Physical Activity Questionnaire (IPAQ). Exclusion criteria were the following: (1) diagnosis of type 2 diabetes; (2) unregulated high blood pressure (over 150 for systolic and 100 for diastolic blood pressure [DBP]); (3) decreased kidney function ($\text{eGFR} > 60\text{ mL/min/1.73m}^2$); (4) glycated haemoglobin > 8.5 ; (5) use of insulin and/or sulphonylureas; (6) engagement in a structured/organised exercise program 6 months prior to the beginning of the study. Written informed consent was obtained before any action in relation to the individuals' participation in the study took place.

Twenty-five participants (15♀, 10♂) met the eligibility criteria, consented to participate and completed all the assessments and experimental conditions. Their characteristics are presented in Table 1.

Assessments

Participants were instructed to fast overnight and refrain from caffeine, alcohol and moderate-to-vigorous physical activity 12 h prior to each visit.

Anthropometry and body composition (Visit 1)

Body mass was measured with participants barefoot and wearing light clothing, using an electronic scale (CPW plus – 300, ADAM Equipment, UK) with 300 kg capacity and 0.1 kg precision. Height was measured using a wall-coupled stadiometer (SECA, Hamburg, Germany) with participants barefoot in an upright position. Body composition was assessed via dual-energy X-ray absorptiometry (DXA) (Lunar iDXA, GE Healthcare, USA). Participants were positioned in the supine position within the designated scanning area. Foam was placed under the armpit to separate the upper limbs from the trunk, and the feet were strapped to avoid lower limb movement. Participants were advised to remain still and keep their eyes closed during the assessment. Total body and segmental indices of lean mass, fat mass, and bone mineral density were acquired from the iDXA analysis.

Cardiorespiratory fitness test (Visit 1)

A bicycle ergometer-graded exercise test was used to determine participants' $\dot{V}O_2\text{max}$ and maximum heart rate (HR_{max}), which was used to determine and monitor exercise intensity for the endurance exercise performance. The adapted protocol started at 50 W, with progressive increments of 25 W every 2-minute stage, until individuals reached volitional exhaustion. Participants were monitored to maintain 50–60 revolutions per minute (rpm) throughout the test. Verbal encouragement was given to stimulate maximal effort. Heart rate (HR; Polar H10 HR monitor, Polar Electro Oy, Finland) and rate of perceived exertion were recorded at the end of every stage. Respiratory measures, including oxygen and carbon dioxide consumption, were continuously measured using a previously calibrated metacart (Parvomedics TrueOne 2400, Salt Lake City, UT).

Echocardiographic assessments

Cardiac function was assessed by means of a transthoracic echocardiogram using a commercially available ultrasound

system (CX50; Philips Ultrasound, Andover, MA, USA) with a S5-1 MHz sector array transducer. A single highly experienced sonographer (JJC), who was blinded regarding group allocation, performed all the echocardiograms according to comprehensive guidelines (Mitchell et al. 2019). Echocardiographic loops were obtained with participants lying in the supine left lateral decubitus position at the end of a normal breath. At least 3 complete cardiac cycles were obtained for every image, with depth minimised to optimise frame rate. Images were digitally stored in cine-loop DICOM format and transferred to an online workstation (TOMTEC, TOMTEC Imaging Systems GmbH, Munich, Germany) for analysis. Echocardiographic assessments were performed immediately before the commencement of each session and repeated no longer than a minute after session completion, taking approximately 3 min to complete, totalling 4 min from protocol conclusion to echocardiographic assessment completion. Upon conclusion of the HWI or END protocol, participants were provided a towel to quickly dry their thoracic area, and were then assisted to lie down in an echocardiographic bed placed next to the cycle ergometer/hot water tank for echocardiographic assessment.

Left ventricular volumes (i.e. LV end-diastolic volume [EDV]; and LV end-systolic volume [ESV]) and function (i.e. LVEF) were defined using 2D imaging, applying Simpson's method of disks. Stroke volume was calculated as the difference between EDV and ESV. Left ventricle GLS was assessed via speckle-tracking echocardiography and quantified using 2D strain images. Images were analysed using the AutoStrain LV package (TOMTEC, TOMTEC Imaging Systems GmbH, Munich, Germany), which is an application that provides measurement tools for 2D strain analysis of the LV. AutoStrain LV follows the Strain Task Force Consensus documents. Mean values from the 18 segments obtained from the apical four-, three-, and two-chamber views were used to calculate global LV GLS.

LV diastolic function assessment included the assessment of diastolic filling waves and myocardial tissue velocity. With the sample volume positioned at the tip of the mitral valve (MV), the peak early (E) and late (A) transmittal inflow velocities were measured and used to calculate the ratio between early and late ventricular filling velocities (E/A). Myocardial tissue velocities were assessed using tissue Doppler imaging. The sample volume was positioned at the septal and lateral annulus for the acquisition of the velocity of the MV annulus tissue in early (e') and late (a') diastole, and peak systolic velocity (s'). The early diastolic velocity ratio (E/e') was calculated using E and e' velocities. The assessments of LV function were performed and analysed in accordance with the procedures established by

Table 1 Participants' characteristics and acute changes in thermoregulation, haemodynamic and cardiac function according to the experimental conditions

Anthropometrics			Total				
n			25		-		-
Age (years)			59±7		-		-
Height (cm)			169±8		-		-
Weight (kg)			84.3±11.6		-		-
BMI (kg/m ²)			29.4±3.4		-		-
BSA (m ²)			1.95±0.16		-		-
<i>Body composition</i>							
Lean mass (kg)			47.8±7.3		-		-
Lean mass (%)			57.4±6.3		-		-
Fat mass (kg)			33.1±8.3		-		-
Fat mass (%)			40.7±6.7		-		-
<i>Cardiac morphology</i>							
IVSd (cm)			1.01±0.13		-		-
LVIDd (cm)			4.58±0.39		-		-
LVPWd (cm)			1.01±0.13		-		-
LVIDs (cm)			2.96±0.52		-		-
LV mass (g)			160.49±38.81		-		-
LV mass/BSA (g/m ²)			81.26±14.86		-		-
END (n=25)			HWI (n=25)		P-value		
	Baseline	Post*	Baseline	Post*	Time	Condition	INT
<i>Thermoregulation</i>							
Tc (°C)	37.0±0.3	37.8±0.3	37.1±0.3	38.1±0.4	<0.001	0.016	0.004
<i>Hemodynamics</i>							
SBP (mmHg)	127±16	125±12	129±13	123±10	0.045	0.648	0.043
DBP (mmHg)	74±10	72±8	73±10	64±10	<0.001	0.013	0.020
HR (bpm)	67±10	87±12	68±14	95±17	<0.001	0.130	0.042
<i>LV Function</i>							
LV GLS (%)	-21.00±2.34	-21.7±2.6	-20.9±2.7	-23.5±2.9	<0.001	0.259	<0.001
Stroke Volume (ml)	53.34±10.21	50.16±10.49	50.26±8.62	47.57±11.08	0.004	0.047	0.895
LV EDV (ml)	84.56±19.99	72.26±16.40	79.42±18.5	67.99±18.27	<0.001	0.037	0.920
LV ESV (ml)	31.23±11.78	22.14±7.41	29.16±10.96	20.42±8.46	<0.001	0.177	0.971
Cardiac Output (ml)	3269±637	4358±1090	3147±656	4456±971	<0.001	0.475	0.403
LV EF (%)	64±7	70±5	64±6	71±5	<0.001	0.960	0.501
MV Pk E (cm/s)	66.1±10.43	69.05±17.34	68.0±14.91	75.67±16.26	0.008	0.142	0.183
MV Pk A (cm/s)	72.2±10.13	78.9±16.54	72.0±13.64	85.48±16.44	<0.001	0.196	0.338
MV E/A ratio	0.93±0.21	0.92±0.39	0.97±0.25	0.9±0.19	0.603	0.860	0.729
Average e' (cm/s)	9.1±1.6	9.8±2.2	9.0±1.8	10.3±1.8	<0.001	0.439	0.144
Average a' (cm/s)	11.8±2.0	13.0±2.4	11.6±2.2	14.3±2.0	<0.001	0.114	0.005
E/e' average	7.35±1.53	7.18±1.55	7.52±1.83	7.54±1.72	0.725	0.349	0.666
e'/a'	0.80±0.23	0.78±0.21	0.82±0.28	0.73±0.18	0.034	0.727	0.111
Average s' (cm/s)	9.18±1.52	12.94±2.27	8.85±1.4	13.99±2.27	<0.001	0.284	0.072
<i>RV Function</i>							
RVFWLS (%)	-25.0±4.2	-24.3±5.3	-26.5±3.9	-26.1±5.2	0.814	0.166	0.156
TAPSE (mm)	22±5	23±4	21±4	25±4	0.009	0.259	0.023

Data are presented as mean±standard deviation. A, mitral inflow blood velocity during atrial contraction; a', mitral annulus tissue velocity during atrial contraction; BMI, body mass index; BSA, body surface area; DBP, diastolic blood pressure; E, mitral inflow early velocity; e', early diastolic mitral annulus velocity; EDV, end-diastolic volume; EF, ejection fraction; END, endurance exercise; ESV, end-systolic volume; GLS, global longitudinal strain; HR, heart rate; HWI, hot water immersion; INT, interaction between time and group; IVSd, interventricular septum diameter; LV, left ventricle; LVIDd, left ventricle internal diameter at end-diastole; LVIDs, left ventricle internal diameter at end-systole; LVPWd, left ventricle posterior wall thickness; MV, mitral valve; Pk, peak; s', systolic mitral annulus tissue velocity; RVFWLS, Right ventricular free wall longitudinal strain; SBP, systolic blood pressure; TAPSE, tricuspid annular plane systolic excursion; Tc, core temperature. Significance was set at $p \leq 0.05$. Bold values indicate significance. * Data collection was completed 4 min after cessation of the experimental conditions

the American Society of Echocardiography (ASE) (Lang et al. 2015).

Right ventricular free wall longitudinal strain (RVFWLS) was assessed using a RV-focused four-chamber view. Mean values from the 3 segments were used to calculate RVFWLS. The AutoStrain RV (TOMTEC, TOMTEC Imaging Systems GmbH, Munich, Germany) package was used to assess RVFWLS. RV systolic function was quantified via the assessment of the tricuspid annular plane systolic excursion (TAPSE), using the M-mode ultrasound technique. The assessments of RVFWLS and RV function strictly followed the recommendations presented in the respective dedicated guidelines (Badano et al. 2018; Mukherjee et al. 2025).

Experimental conditions

Participants who met the eligibility criteria underwent two experimental conditions (i.e. HWI and END), performed in a random order, at the same time of the day and separated by at least 72 h, wherein the ambient temperature averaged 24 °C for both conditions (Fig. 1). Participants were asked to refrain from caffeine, alcohol and any moderate-to-vigorous

physical activity for 24 h prior to experimental condition performance.

Hot water immersion (HWI)

With the water preset at 40 °C, participants were instructed to sit inside the hot water tank with the water up to the manubrium, keeping their arms under the water for the whole session. Participants were asked to ingest a radiotelemetry gastrointestinal pill (eCelsius Performance; Body-Cap Medical, Hérouville-Saint-Clair, France) 6 h prior to the commencement of the session to have their body core temperature (T_{c}) monitored throughout the session. Core temperature was measured every 10 min during the performance of the experimental condition. Blood pressure was measured immediately before and no later than 4 min after the completion of the session using a validated electronic blood pressure monitor (Dinamap® Carescape V100, GE Healthcare, California, USA). HR was assessed every 10 min using a HR monitor connected to a chest strap (Polar H10 HR monitor, Polar Electro Oy, Finland). Participants were also monitored for heat-related symptoms every 10 min and asked to answer yes or no to a series of questions pertaining

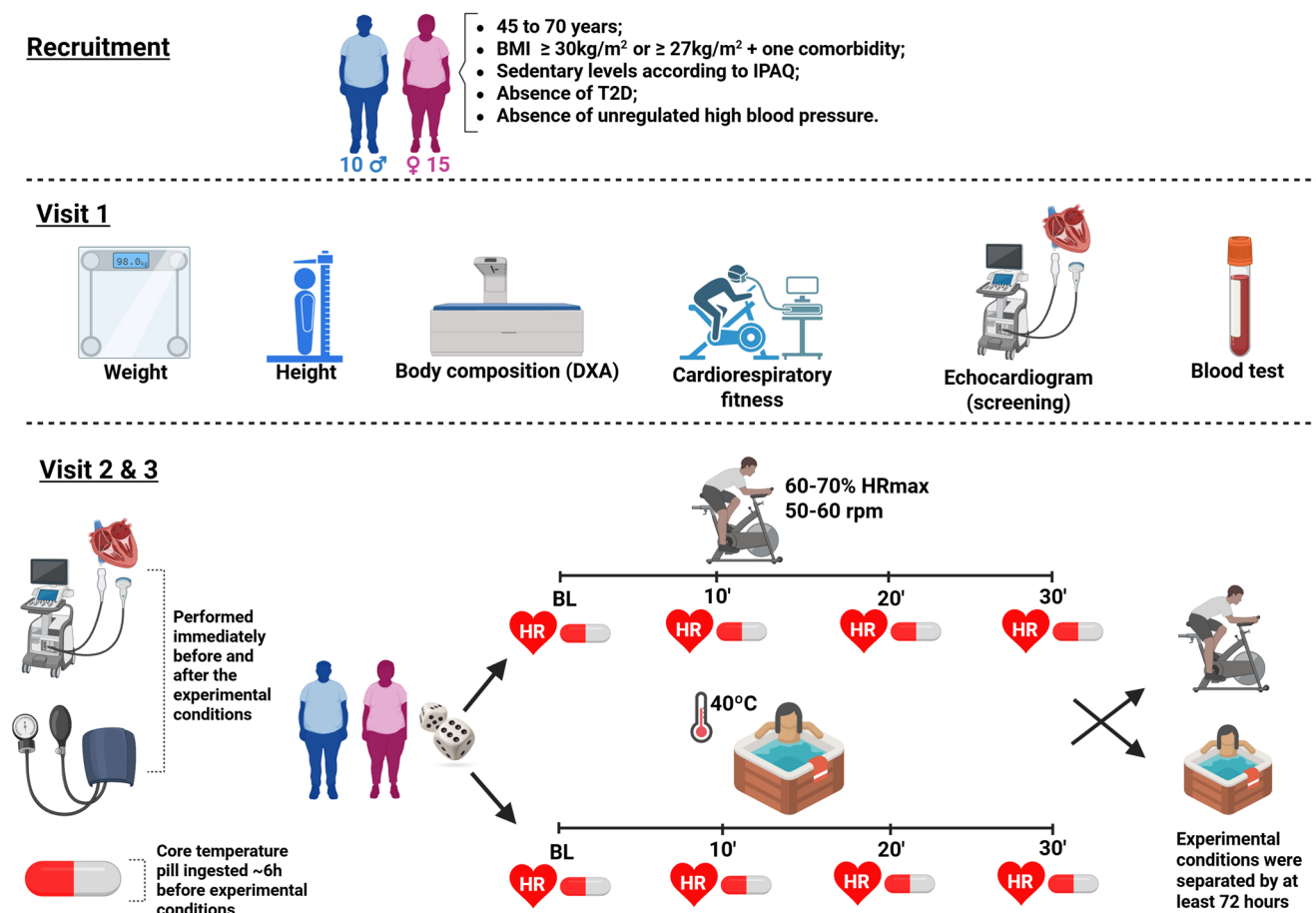


Fig. 1 Illustration of study design. Created with BioRender.com

to heat-related symptoms (e.g. lightheadedness, heart beating fast, weakness, breathlessness, headache, dizziness). If they say yes to any of the questions, they were immediately assisted to exit the hot water tank or stop exercise, being closely monitored until symptom dissipation. During this time, blood pressure measurements were taken to guarantee hemodynamic homeostasis and participants' safety.

Endurance exercise session

Participants underwent a cycling-based exercise session of 30 min with intensity kept at 65–70 HR_{max}. Participants were advised to maintain cycling cadence at 50–60 rpm. Blood pressure was measured before and no later than 4 min after the completion of the exercise session. Tc and HR were assessed every 10 min using the same equipment and protocol described above.

Statistical analysis

Continuous variables are expressed as mean and standard deviation. Data were analysed using SPSS (version 29.0) (IBM Australia Ltd, Australia). Data normality was checked via the Shapiro-Wilk test. Differences between experimental conditions over time were analysed using 2-way repeated measures ANOVA. In the event that statistical significance was found, a paired samples t-test was conducted to compare differences in time points within the same group, and an independent samples t-test was conducted to analyse differences between conditions. The significance level was set at $P \leq 0.05$.

Results

Participants' characteristics and changes according to the experimental conditions are presented in Table 1. There was no significant difference at baseline between experimental conditions for any of the variables included in the study.

Core temperature and hemodynamic

Tc gradually increased throughout both experimental conditions, with a significantly greater increase during HWI (interaction: $p=0.004$; condition: $p=0.016$; time: $p<0.001$) (Table 1). A significant interaction and time effect were found for HR ($p=0.042$ and $p<0.001$, respectively). Despite the fact that both END and HWI induced a significant increase in HR ($p<0.001$; $p<0.001$), HWI promoted a significantly greater increase in HR ($p=0.042$) (Fig. 2B). Significant interaction ($p=0.043$) and time effects ($p=0.045$) were found for SBP. Whilst END induced no change in SBP,

HWI induced a significant decrease in this variable (HWI: $\Delta = -6 \pm 10$ mmHg) (Fig. 2C). Regarding DBP, significant interaction ($P=0.020$), time ($P<0.001$), and condition ($P=0.013$) effects were also observed. DBP was slightly reduced after END ($\Delta = -2 \pm 5$ mmHg), but greatly reduced after HWI ($\Delta = -9 \pm 9$ mmHg) (Fig. 2D). There was a significant difference in changes in SBP ($P=0.019$) and DBP ($P=0.001$) after HWI in comparison with changes after END.

Changes in left ventricular function

LV deformation and volumes are shown in Fig. 3. Significant time ($p<0.001$) and interaction ($p<0.001$) effects were found for LV GLS (Table 1), with no condition effect ($p=0.259$). The augmentation in LV GLS (i.e. more negative) was significantly higher in comparison with the augmentation after END ($p=0.008$) (Fig. 3A). There was a significant decrease in SV (time: $p=0.004$) after both experimental conditions (condition: $p=0.047$), with no significant interaction effect ($p=0.895$) (Table 1). There was no difference in changes in SV after END ($\Delta = -3.17 \pm 6.54$) or HWI ($\Delta = -2.82 \pm 7.81$) ($p=0.866$) (Fig. 3B). There was a significant decrease in EDV following both END and HWI with significant time ($p<0.001$) and condition effects ($p=0.037$), but no interaction between them ($p=0.920$) (Table 1). Change in EDV after END was similar to the change after HWI ($p=0.839$) (Fig. 3C). ESV was significantly reduced after both END and HWI with time ($p<0.001$), but no condition ($p=0.177$) or interaction effects ($p=0.971$) (Table 1), with no significant differences between conditions (Fig. 3D). Cardiac output and EF increased significantly after END and HWI (time: $p<0.001$; $p<0.001$), with no significant condition ($p=0.475$; $p=0.960$) or interaction ($p=0.403$; $p=0.501$) effects (Table 1).

Transmitral flow and myocardial velocities are presented in Table 1. Regarding diastolic variables, time effects were observed for E (time: $p=0.008$), A (time: $p<0.001$), and average e' (time: $p<0.001$) were found for END and HWI. Significant increases in E ($p=0.010$) and A ($p<0.001$) were observed after HWI, but not after END ($p=0.316$; $p=0.107$, respectively). Average e' significantly increased after both HWI ($p<0.001$) and END ($p=0.019$) compared to baseline. A time effect was found for e'/a' ratio. However, HWI induced a significantly greater decrease in this variable in comparison with END (HWI $\Delta = -0.11 \pm 0.19$ vs. END $\Delta = 0.0002 \pm 0.16$; $p=0.039$); this variable only significantly decreased after HWI ($p=0.012$) in comparison to baseline values. A significant time ($p<0.001$) and interaction ($p=0.005$) effect was found for average a' , in which HWI promoted a significantly greater increase in average a' than END (HWI $\Delta = 2.87 \pm 2.15$ vs. END $\Delta = 1.09 \pm 1.23$;

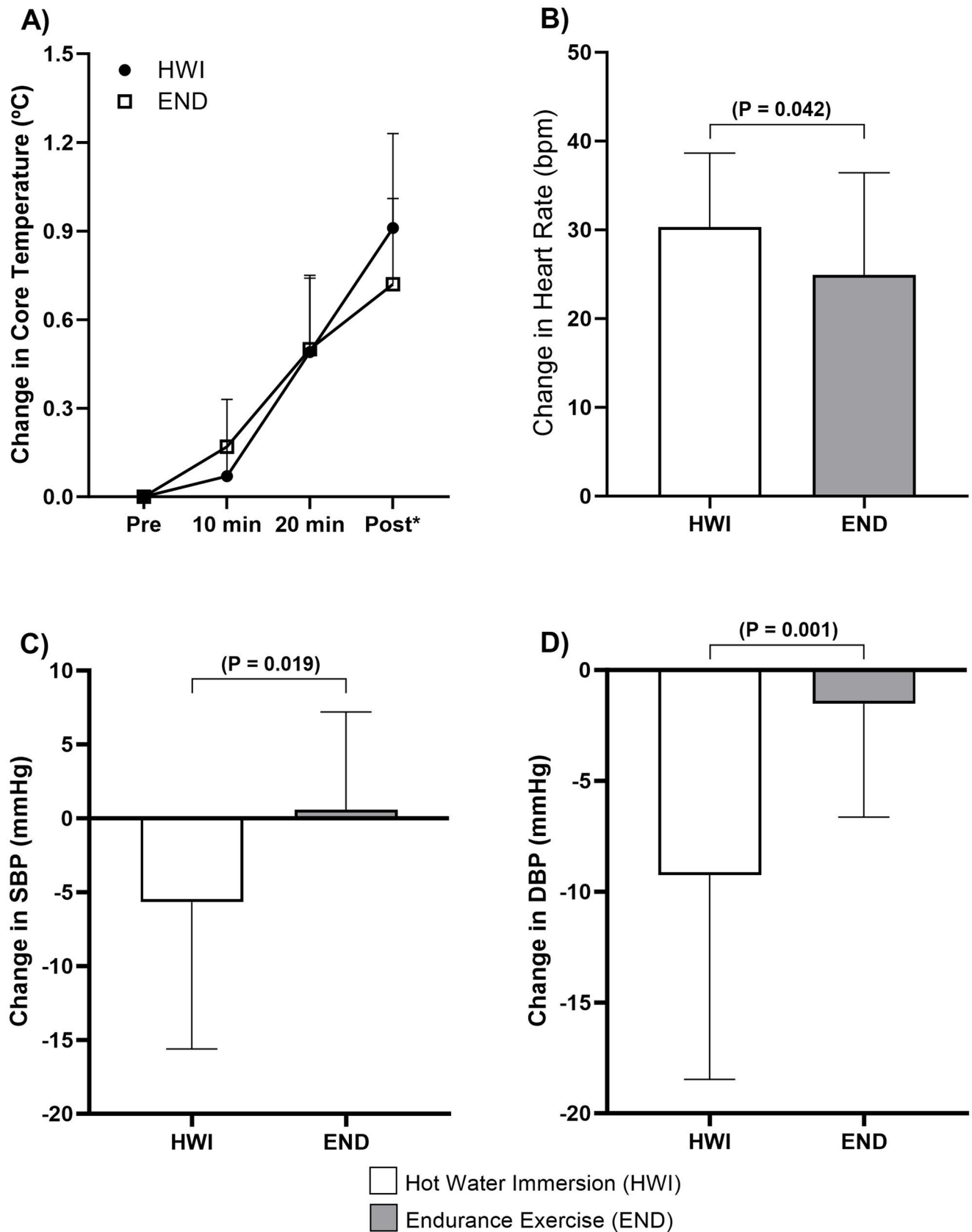
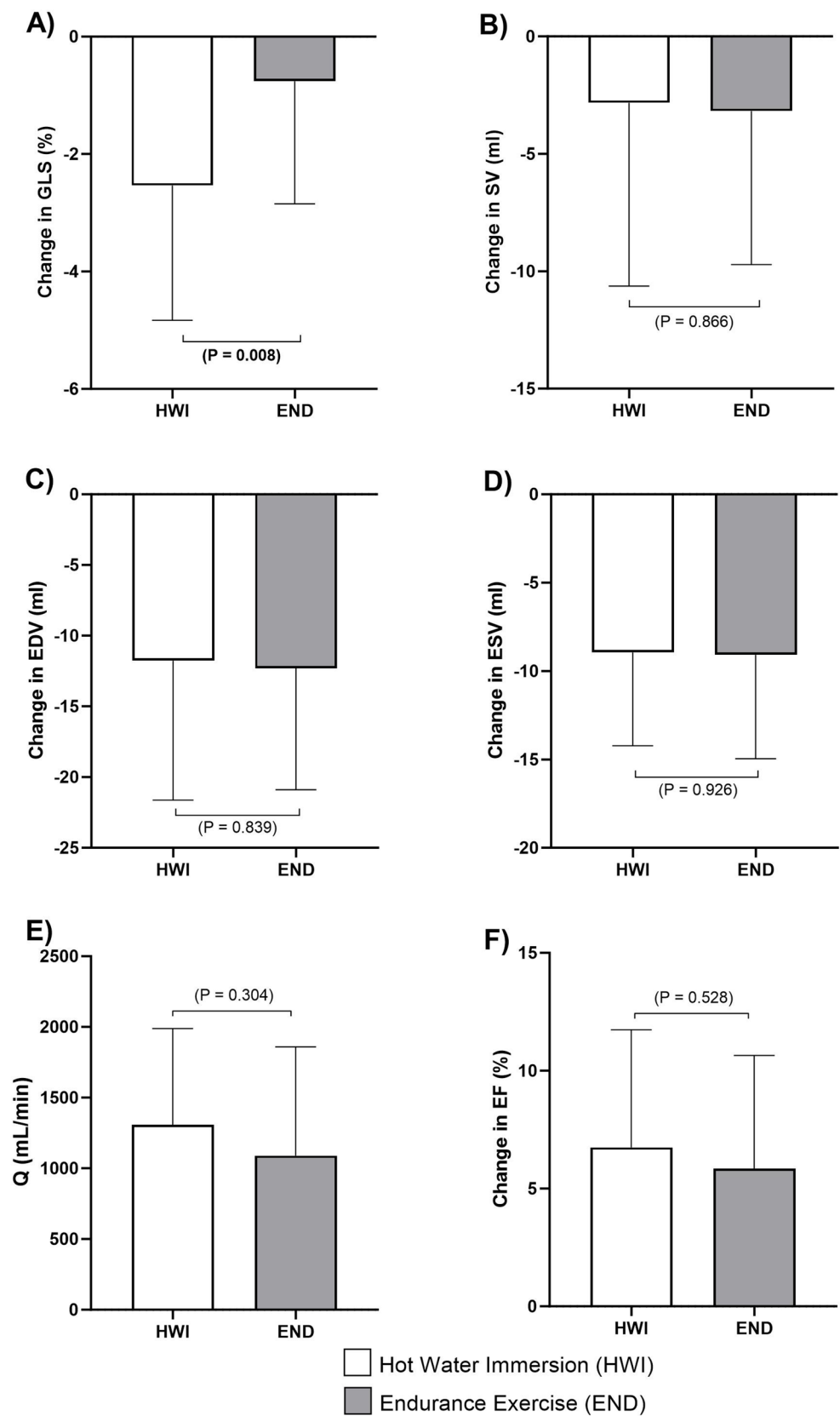


Fig. 2 Changes in (A) core temperature, (B) heart rate, (C) systolic blood pressure and (D) diastolic blood pressure in response to each experimental condition. DBP, diastolic blood pressure; HR, heart rate;

HWI, hot water immersion; SBP, systolic blood pressure. *Data collection was completed 4 min after cessation of experimental conditions

Fig. 3 Left ventricular volumes and deformation before and immediately after the experimental conditions. Significance was set at $p \leq 0.05$. EDV, end-diastolic volume; END, Endurance exercise; ESV, end-systolic volume; HWI, hot water immersion; LV GLS, left ventricular global longitudinal strain; Q, cardiac output. Bold values indicate significance



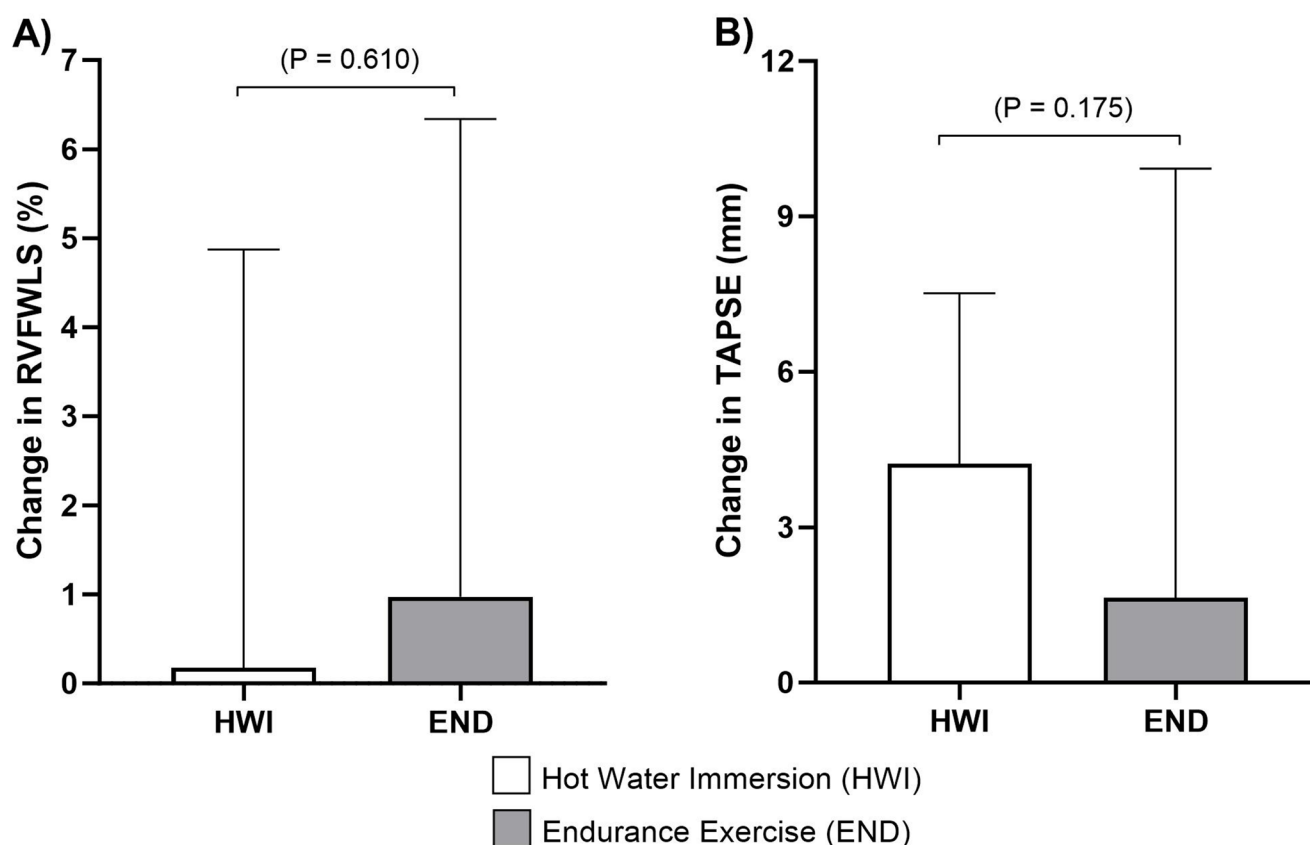


Fig. 4 Right ventricular function before and immediately after the experimental conditions. END, endurance exercise; HWI, hot water immersion; RVFWLS, right ventricular free wall longitudinal strain; TAPSE, tricuspid annular plane systolic excursion. Bold values indicate significance

$p=0.001$). Concerning systolic variables, a time effect was observed for average s' (time: $p<0.001$); an increase in s' was observed after both HWI ($p<0.001$) and END ($p<0.001$). No interaction was reported for the other transmitral flow and myocardial velocity parameters ($p>0.05$).

Changes in right ventricular function

Changes in RV function after both experimental conditions are shown in Table 1. There was no time ($p=0.814$), condition ($p=0.166$) or interaction ($p=0.156$) effect on RVFWLS, with no differences in change in RVFWLS between groups ($p=0.610$) (Fig. 4A). Time ($p=0.009$) and interaction effects ($p=0.023$) were found for TAPSE, respectively. Only HWI significantly increased TAPSE ($p<0.001$), but there was no significant difference in changes in TAPSE between conditions ($p=0.175$) (Fig. 4B).

Discussion

In the present study, we investigated the acute effects of HWI and END on cardiac function in overweight middle-aged individuals. We found that: (1) HWI and END induced

similar decreases in EDV, ESV, and SV, alongside increases in HR, EF and Q. However, only HWI significantly increased LV GLS, while the increase in LV GLS after END was not significant; (2) HWI induced significantly greater increases in Tc and HR, with more pronounced decreases in SBP and DBP in comparison with END; (3) HWI induced significantly greater increases in RV systolic function than END, with a significant increase in TAPSE post intervention, whereas END showed a modest non-significant increase in this variable.

In our study, we found that LV GLS only increased with HWI. At the same time, HWI induced significantly greater reductions in SBP and DBP in comparison with END. Given that lower blood pressure leads to decreased afterload, this may have mediated the change in LV GLS. In fact, previous evidence indicates that reduced LV GLS is associated with high blood pressure (Bendiab et al. 2017). A 10-year follow-up study demonstrated that LV GLS was significantly reduced in hypertensive patients, being associated with major adverse cardiac events (MACE), regardless of other clinical and echocardiographic parameters (Tadic et al. 2024). Other studies have found similar outcomes in relation to changes in blood pressure. Roxburgh and colleagues (2023) observed a reduction of 12 mmHg and 7

mmHg in SBP and DBP, respectively, after a bout of HWI in individuals with severe lower limb osteoarthritis. Furthermore, lower-limb HWI (30 min, 42 °C) reduced both SBP (-10 mmHg) and DBP (-6 mmHg) in 10 healthy young adults, whereas no changes in blood pressure were observed after END (treadmill running at 65–75% of age-predicted HR_{max}). Likewise, our study showed a decrease in SBP and DBP of 6 mmHg and 9 mmHg, respectively, with no significant changes after END. Given that LV volumes decreased after HWI, we could not ascribe changes in LV GLS to preload augmentation or the Frank-Starling mechanism. This emphasises the key role of blood pressure in reducing afterload and consequently augmenting LV GLS.

Regarding changes in LV volumes after HWI and exercise, a study performed in 12 healthy young adults who were exposed to a session of HWI (40.5°C, 60 min) and exercise (60% of VO₂max, 60 min) reported that both HWI and END induced a decrease in SV, alongside increases in HR and Q. Nevertheless, END promoted a significantly greater increase in Q and HR, combined with a significantly lower SV than HWI (Francisco et al. 2021). Similarly, Thomas and colleagues (2016) found that a session of END (treadmill running at 65–75% of age-predicted HR_{max}) increased HR and Q to a higher degree than lower-limb HWI (42°C, 30 min) in 10 healthy young adults. In our study, both HWI and END induced similar decreases in EDV, ESV and SV, combined with increases in EF and Q. These findings may have occurred in the context of a significant increase in HR after both HWI (+33 bpm) and END (+25 bpm). Increased HR can be associated with a reduced ventricular filling time, which leads to a reduced EDV and SV, while also increasing Q (Kagami et al. 2022). Our findings differ from previous studies, as no significant differences were observed regarding changes in LV volumes between HWI and END in our study. This may be ascribed to the exercise protocol used, the level of the water during HWI, but most importantly, to the population we studied. Whilst the majority of the studies investigated healthy young adults, our study included middle-aged individuals with overweight/obesity. Given that both aging (Millyard et al. 2020) and excess fat mass (Speakman 2018) impact thermoregulation and cardiac function, HWI might have exerted a more pronounced impact in this population than that observed in previous studies. Our study also showed that, despite both HWI and END promoting changes in systolic and diastolic parameters of LV tissue velocity, only HWI significantly increased peak E, peak A, and e'/a' ratio. These findings corroborate previous evidence that demonstrated an increase in peak E, A and s' following a HWI protocol in both healthy and spinal cord-injured individuals (Kouda et al. 2024). This highlights the capacity of HWI protocols to induce changes in LV function in older and diseased populations.

To our knowledge, the present study is the first to investigate and compare the acute effects of HWI and END on RV function. We demonstrated that a session of HWI promoted greater increases in RV systolic function (i.e. increase in TAPSE). Our findings are in accordance with the only other evidence available in the literature, which found an increase in RV tissue velocity after a session of warm water immersion (34 °C for 20–30 min) in patients with chronic heart failure. We speculate that the increase in RV function after HWI is associated with a reduction in total peripheral resistance caused by an increase in Tc. Elevation in skin and core temperature triggers thermoregulatory mechanisms that induce skin vasodilation, which in turn decreases ventricular filling pressure and end-diastolic volume. At the same time, compensatory mechanisms maintain adequate blood supply by increasing RV systolic and diastolic function (Chou and Coyle 2023). That being said, the greater increase in Tc and reduction in blood pressure appear to be responsible for an increase in RV function after HWI, but not after END. Given the increasingly important role of RV function as a prognostic marker of cardiovascular morbidity and mortality (Amsallem et al. 2018), these findings shed light on HWI as a potential treatment alternative able to induce significant changes in both LV and RV function. In spite of HWI-induced increases in some parameters of RV function, no significant change in RVFWLS was observed after HWI and END. Right ventricular free wall longitudinal strain is a more reproducible parameter of RV function than TAPSE, with the ability to detect subclinical RV dysfunction (Muraru et al. 2022). Right ventricular free wall longitudinal strain is also the echocardiographic parameter that best correlates with the gold-standard MRI-derived right ventricular ejection fraction (Agasthi et al. 2020). The findings related to changes in RV function after HWI need to be interpreted with caution, as more studies are warranted to investigate the acute effects of HWI and END on more reproducible parameters of RV function.

The present study has several limitations. We did not match the increase in Tc during the performance of the experimental conditions, where HWI induced a greater increase in Tc than END. We aimed to conduct an ecologically valid experiment where experimental conditions were time-matched. Therefore, the greater acute changes induced by HWI on cardiac function might be attributed, in part, to a greater increase in Tc. Another limitation relates to the timing of our cardiac assessments. Whilst there are no current methodological approaches that facilitate real time contemporaneous assessment of cardiac functional outcomes in the postures and physical constraints associated with acute HWI or exercise interventions, we collected post hoc measurements within 4 min of moving to the echocardiography bed. Another question relates to the persistence of the effects of

HWI and END intervention. Cardiac responses during the exercise intervention are greater than during the HWI protocol, but these differences may diminish more rapidly during recovery, whereas the heating effect may persist for a longer period. Future studies should investigate cardiac responses repeatedly across a prolonged recovery period. Also, many of the individuals included in the study were on continuous medication use for diverse comorbidities (Supplementary Table 1), and this may have impacted the results of the present study to some extent. Another limitation is the impact of posture on the cardiac-related responses that we assessed. We performed echocardiographic assessment in a semi-recumbent position to optimise image quality, whereas participants were seated during both HWI and END protocols. This may have impacted our outcomes related to cardiac function. Nevertheless, our assessments are more ecologically valid than those collected in MRI-based experiments, where participants are placed in a fully recumbent position and have their breathing patterns manipulated. In addition, although our post-intervention echocardiographic assessment was completed approximately 4 min after participants exited the cycle ergometer or hot water tank, previous evidence indicates that systolic and diastolic echocardiographic parameters remained relatively altered 5 min after exercise performance (Locatelli et al. 2024). This suggests that the less than 5-minute interval between experimental condition cessation and completion of echocardiographic assessment had minimal impact on our results. Finally, we acknowledge the importance of investigating sex differences in physiological studies, and future experiments might involve sub-groups that are deliberately matched for variables such as BSA or body composition. The present study was underpowered for such analysis and also for any correlational analyses, as we focused more generally on the main findings. In future, larger studies may address questions of association.

Hot water immersion induced significantly greater increases in both LV and RV function parameters in comparison to time-matched END, highlighting the potential of HWI to promote changes in cardiac function and its potential as a nonpharmacological treatment for the improvement of cardiac function. In addition, the findings from the present study shed light not only on the potential of HWI to induce acute changes in traditional and novel echocardiographic indices of LV function but also provide important insights about its acute impact on RV function, an often overlooked but increasingly important tool in clinical practice. Future studies are warranted to investigate whether the acute effects observed in our study could translate into chronic adaptation and if there is a difference in chronic adaptations in cardiac function between HWI and END in this population.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s00421-026-06418-z>.

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Funding Open Access funding enabled and organized by CAUL and its Member Institutions. J.C.L. was supported by a Co-Funded University of Western Australia Postgraduate Award. D.G. and L.H. were supported by the National Heart Foundation – Testing the feasibility of using heat therapy as a new treatment for patients with advanced heart failure. H.J., K.G. and D.G. were supported by the International Investment Initiative (i3): i-CARDIO – International Collaboration Assessing novel health models: A Research Design Intervention Opportunity.

Data availability The data underlying this article will be shared on reasonable request to the corresponding author.

Declarations

Conflict of interest The authors declared that they have no conflict of interest.

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